

THÈSE DE DOCTORAT

Titre français

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du grade de docteur en** Computer Science
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Tritre francais

xi+22 p.

*À toi lecteur <3 :**

Titre français

Mots-clés : Pâtisserie, Macaron, Ladurée.

English Title

Abstract

Macarons are so good.

Keywords: Pastry, Macaron, Ladurée.

Remerciements

Merci !

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CHAPTER 1

Introduction

1.1 Prologue

Automation everywhere in CS, we want to set up boilerplates to help our development.

Examples:

- Continuous integration
- Macros are everywhere
- Artificial intelligence

We delegate tasks to some tools.

expert system →

example plus frappant:

artificial intelligence: based on probability but not reproducible, not predicatable

several other examples in computer sciences:

...

1.2 Automation is everywhere

In the domain of formal proof automation is everywhere, and it is formally called elaboration ??

In formal proof, proofs are verbose and have lot of details: types, coercions, canonical structures, type-classes.

In theorem provers the user benefit of notations, implicit inference and tactics to ease his job.

1.2.1 Elaboration in theorem provers

Simply typed lambda calculus is an example of automation: an algo checks if ...

1.2.2 Type classes

1.3 Contributions

In this manuscript we focus on an alternative type-class solver written in the elpi meta-language.

We also talk about a formalized tool. . .

CHAPTER 2

State of the Art

2.1 Meta-programming

Tactics, coq-elpi

```
func p int -> int.
```


CHAPTER 3

Type classes resolution in Elpi

Discrimination trees: opt for lists and tree depths.

The entry point. . .

[\[1\]](#)

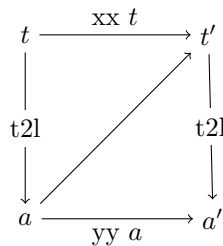
[\[2\]](#)

CHAPTER 4

Determinacy checking for Elpi

hard cut as in prolog

[3]



Discrimination trees: discriminate rules head

Inputs must come before outputs, with Disc tree this is not mandatory, since unification is weaker.

EG:

```
pred g o:int, o:int.
```

```
f 3 3 (g X X).
```

```
f 3 4 (g 3 4).
```

```
main:-
```

```
f X Y (g X Y).
```

In the code above, both rules can be used with the query, but, the discrimination tree, save the world

CHAPTER 5

Conclusion

Les macarons c'est bon et ceux de Ladurée sont semble-t-il meilleurs. [?]

5.1 Perspectives

Encore beaucoup de travail à faire.

Définition 5.1.1. TEST

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- [3] Davide Fissore and Enrico Tassi. Determinacy checking for elpi: an higher-order logic programming language with cut. In Nada Amin and Joaquín Arias, editors, *Practical Aspects of Declarative Languages*, pages 77–95, Cham, 2026. Springer Nature Switzerland.

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Theorem's list

Appendix

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BLIBLI

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